Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN Stack

M2:Practical – IX

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9.User Data Management
a. Create an API to add user details using Express and MongoDB.
b. Design a React form to send user data to the backend.
c. Retrieve and display user data from MongoDB.
d. Implement an update operation for existing records.
e. Implement the delete operation for selected records.
f. Display updated records dynamically in React.

Code:
server.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");

const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// MongoDB Connection
mongoose.connect("mongodb://127.0.0.1:27017/UserDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed");
    console.log(error);
  });
// GET - Retrieve Users
app.get("/users", async (req, res) => {
  try {
    const users = await mongoose.connection.db
      .collection("users")
      .find()
      .toArray();
    res.json(users);
  }
  catch (error) {
    res.status(500).json({ message: "Error fetching users" });
  }
});
// POST - Add User
app.post("/users", async (req, res) => {
  try {
    const user = req.body;
    await mongoose.connection.db
      .collection("users")
      .insertOne(user);

    res.json({ message: "User added successfully" });
  }
  catch (error) {
    res.status(500).json({ message: "Error adding user" });
  }
});
// PUT - Update User
app.put("/users/:id", async (req, res) => {
  try {
    const id = new mongoose.Types.ObjectId(req.params.id);
    const updatedUser = req.body;
    await mongoose.connection.db
      .collection("users")
      .updateOne({ _id: id },
        {
          $set: {
            name: updatedUser.name,
            email: updatedUser.email,
            city: updatedUser.city
          }
        }
      );
```

```
        res.json({ message: "User updated successfully"
});
    }
    catch (error) {
        res.status(500).json({ message: "Error updating
user" });
    }
});
// DELETE - Delete User
app.delete("/users/:id", async (req, res) => {
    try {
        const id = new
mongoose.Types.ObjectId(req.params.id);
        await mongoose.connection.db
            .collection("users")
            .deleteOne({
                _id: id

            });
        res.json({ message: "User deleted successfully"
});
    }
    catch (error) {
        res.status(500).json({ message: "Error deleting
user" });
    }
});
// Start Server
app.listen(PORT, () => {
    console.log(`Server is running on
http://localhost:${PORT}`);
});
```

App.jsx:

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
  // Store all users
  const [users, setUsers] = useState([]);
  // Form data
  const [name, setName] = useState("");
  const [email, setEmail] = useState("");
  const [city, setCity] = useState("");
  // Store ID of user being edited
  const [editId, setEditId] = useState(null);
  // Get Users from MongoDB
  const getUsers = () => {
    axios.get("http://localhost:3000/users")
      .then((response) => {

        setUsers(response.data);
      })
      .catch((error) => {
        console.log(error);
      });
  };
  // Load users when page starts
  useEffect(() => { getUsers(); }, []);
  // Add or Update User
  const saveUser = () => {
    const user = {
      name: name,
      email: email,
      city: city
    };
    // UPDATE existing user
    if (editId) {
      axios.put(`http://localhost:3000/users/${editId}`,
user)
        .then((response) => {
          alert(response.data.message);
          clearForm();
          getUsers();
        })
        .catch((error) => { console.log(error); });
    }
    // ADD new user
    else {
      axios.post("http://localhost:3000/users", user)
      .then((response) => {
        alert(response.data.message);
        clearForm();
        getUsers();
      })
      .catch((error) => { console.log(error); });
    }
  };
  // Edit User
  const editUser = (user) => {
    setEditId(user._id);

    setName(user.name);
    setEmail(user.email);
    setCity(user.city);
  };
  // Delete User
  const deleteUser = (id) => {
    if (window.confirm("Are you sure you want to delete
this user?")) {
      axios.delete(`http://localhost:3000/users/${id}`)
        .then((response) => {
          alert(response.data.message);
          getUsers();
        })
        .catch((error) => { console.log(error); });
    }
  };
  // Clear Form
```

```
 const clearForm = () => {
   setName("");
   setEmail("");
   setCity("");
   setEditId(null);
 };
 return (
   <div>
     <h1>User Management System</h1>
     {/* User Form */}
     <h2>
       {editId ? "Update User" : "Add User"}
     </h2>
     <input type="text"
       placeholder="Enter Name"
       value={name}
       onChange={(e) => setName(e.target.value)}
     />
     <br /><br />

     <input type="email"
       placeholder="Enter Email"
       value={email}
       onChange={(e) => setEmail(e.target.value)}
     />
     <br /><br />
     <input type="text"
       placeholder="Enter City"
       value={city}
       onChange={(e) => setCity(e.target.value)}
     />
     <br /><br />
     <button onClick={saveUser}>
       {editId ? "Update User" : "Add User"}
     </button>
     {editId && (<button onClick={clearForm}>
       Cancel
     </button>
     )}
     <hr />
     {/* Display Users */}
     <h2>User List</h2>
     <table border="1">
       <thead>
         <tr>
           <th>Name</th>
           <th>Email</th>
           <th>City</th>
           <th>Actions</th>
         </tr>
       </thead>
       <tbody>

         {users.map((user) => (
           <tr key={user._id}>
             <td>{user.name}</td>
             <td>{user.email}</td>
             <td>{user.city}</td>
             <td>
               <button onClick={() => editUser(user)}>
                 Edit
               </button>
               <button onClick={() =>
deleteUser(user._id)}>
                 Delete
               </button>
             </td>
           </tr>
         ))}
       </tbody>
     </table>
   </div>
 );
}
export default App;
```

Output:

```
> use("UserDB")
< switched to db UserDB
> db.users.insertMany([
    {
      name: "Asmitha",
      email: "asmi@gmail.com",
      city: "Mumbai"
    },
    {
      name: "Anu",
      email: "anu@gmail.com",
      city: "Pune"
    }
  ])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6abcc2dc7b0dba0300bac8e0'),
      '1': ObjectId('6abcc2dc7b0dba0300bac8e1')
    }
  }
```

```
db.users.find()
{
  _id: ObjectId('6abcc2dc7b0dba0300bac8e0'),
  name: 'Asmitha',
  email: 'asmi@gmail.com',
  city: 'Mumbai'
}
{
  _id: ObjectId('6abcc2dc7b0dba0300bac8e1'),
  name: 'Anu',
  email: 'anu@gmail.com',
  city: 'Pune'
}
```









